

Project Summary

The objective of this project was to build a machine learning model capable of predicting house prices based on various property features. The dataset contained information such as area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and other amenities that may influence the value of a house. The project began with data loading and exploration using Pandas. The dataset contained 545 records and 13 columns. During the initial analysis, no missing values were found in any column, and no duplicate records were present in the dataset. This helped simplify the data cleaning process.

Data Preparation

Several categorical columns containing values such as "yes" and "no" were converted into numerical format using one-hot encoding. After preprocessing, the dataset contained 14 columns, including the target variable (price) and the encoded feature columns. The data was then divided into training and testing sets using an 80:20 ratio.

Model Development and Evaluation

Two regression models were trained and evaluated:

Linear Regression

- MAE: 970,043
- RMSE: 1,324,507
- R^2 Score: 0.653

Random Forest Regressor

- MAE: 1,021,546
- RMSE: 1,400,566
- R^2 Score: 0.612

Key Findings

- The Linear Regression model outperformed the Random Forest Regressor on this dataset.
- Linear Regression achieved an R^2 Score of 0.653.
- The Random Forest model achieved an R^2 Score of 0.612.
- Area was identified as the most important feature affecting house prices, followed by Bathrooms, Air Conditioning, Parking, and Stories.

The visualizations further supported these findings. The house price distribution showed variation across different price ranges, while the correlation heatmap highlighted the relationship between house price and several property features. The average price by bedrooms chart indicated that houses with more bedrooms generally tend to have higher prices.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. Through data preprocessing, visualization, model training, and evaluation, meaningful insights were obtained from the dataset. Among the models tested, Linear Regression produced the best results and provided a reliable baseline for predicting house prices based on property characteristics.